KYLE DAWSON, PHD

Lead Data Scientist | Machine Learning, Uncertainty Modeling, Climate & Geospatial Analytics

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EXPERIENCE

Lead Data Scientist — Energy

Centerpoint Energy

Apr 2026 – Present Houston, TX

- lead a new geospatial intelligence (GEOINT) team building and deploying machine learning and geospatial analytics for grid resilience, asset management, and emissions monitoring
- productionized a risk-based vegetation management model (LightGBM outage-risk scoring, LiDAR features, budget-constrained optimization on GCP BigQuery) maximizing reliability improvement (SAIFI) per dollar of tree-trimming spend
- developed GPU-accelerated LiDAR point-cloud processing (PyTorch) to map transmission infrastructure and register towers against as-built assets
- built out a team intranet/documentation site to centralize GEOINT data products, pipelines, and analytics for cross-functional stakeholders

Senior Data Scientist — Climate Tech

SeekOps Inc.

Mar 2022 – Apr 2026 Austin, TX

- designed, deployed, and maintained customer-facing analytics platforms (Plotly Dash on Azure and Android) delivering real-time greenhouse gas quantification, QA/QC, and operational decision support securing contracts between $50-150k.
- built and productionized machine learning models to forecast survey quality, detect anomalies, and constrain uncertainty in methane and CO_2 emission estimates
- implemented Bayesian uncertainty propagation and boundary-layer physical models to generate decision-grade emissions metrics used in regulatory and enterprise reporting
- authored technical documentation, uncertainty assessments, and customer-facing deliverables supporting multi-year enterprise contracts across 10+ countries

Data Scientist — Ag Tech

Bayer CropScience

Feb 2020 – Mar 2022 St. Louis, MO

- applied machine learning and geospatial analytics to UAV-derived imagery for large-scale digital phenotyping and crop performance assessment
- trained and evaluated deep learning object detection models (Detectron2) for tree species classification from high-resolution aerial imagery
- built end-to-end analytics workflows combining relational databases, geospatial data, and AWS S3 into machine-learning-ready datasets, with interactive Plotly Dash apps delivering decision support to stakeholders

Postdoctoral Fellow

NASA Langley Research Center

Oct 2017 – Feb 2020 Hampton, VA

- developed uncertainty-aware aerosol retrieval models from HSRL observations and delivered peer-reviewed results for climate applications

EDUCATION

PhD, Atmospheric Science

North Carolina State University

Graduated 2017 Raleigh, NC

Contributions to the Understanding of Aerosol Microphysics Towards Improving the Assessment of Climate Radiative Forcing

EXECUTIVE SUMMARY

Lead Data Scientist with 8+ years of experience building and deploying machine learning, uncertainty quantification, and data analytics systems across climate tech, geospatial analytics, and applied remote sensing. Proven track record of delivering production-grade models, decision-support tools, and scalable data pipelines used by enterprise customers, scientists, and executive stakeholders. Expert in translating complex physical and statistical models into actionable, customer-facing products deployed in cloud environments.

CORE COMPETENCIES

Stakeholder Communication · Executive Briefings · Technical Leadership · Cross-Functional Collaboration · Decision Support Systems · Product-Oriented Data Science · Requirements Translation · Technical Documentation · Data Storytelling · Model Validation & QA/QC · Uncertainty-Aware Decision Making

DOMAIN EXPERTISE

Machine Learning · Statistical Modeling · Bayesian Inference · Uncertainty Quantification · Regression Analysis · Geospatial Analytics · Remote Sensing · LiDAR & Satellite Imagery · Time Series Analysis · Anomaly Detection · Spatial Statistics · Environmental & Climate Data

TOOLS & TECHNOLOGIES

Python · SQL · Git · JupyterLab · Pandas · NumPy · SciPy · Scikit-Learn · Matplotlib · Seaborn · Plotly · AWS S3 · AWS Redshift · Domino Data Lab · GDAL · QGIS · MATLAB · Spotfire · Azure · Docker · APIs

AWARDS

NASA Postdoctoral Program Fellow

NASA Langley Research Center

Oct 2018 Hampton, VA

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Langley Aerospace Research Scholar

NASA Langley Research Center

Jun 2013 Hampton, VA

PUBLICATIONS

Google Scholar Profile